

# AI in Marketing Capstone

Literature Review, Data Research, and Technology Review Submission Guidelines

## OnBrand AdCopy

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## Preparing Your Literature Review

### 1. Introduction

- This project addresses the challenge of generating high-performing digital advertising copy that simultaneously remains faithful to a brand's established voice and tone. Digital advertising teams routinely produce dozens of ad copy variants and test them through A/B or multivariate experiments, a process that is both time-consuming and expensive. More critically, predicting which text variant will yield a higher click-through rate (CTR) or conversion rate before the campaign goes live remains largely guesswork. The problem is compounded by a second, often conflicting objective: performance-optimized copy frequently drifts from the brand's approved tone and messaging guidelines. Companies attempt to manage this deviation through editorial guidelines and manual review, but these measures are subjective, slow, and not systematically auditable. According to the McKinsey Global Institute (2023), marketing and sales is one of the four functional areas where generative AI could deliver the most value, estimating a productivity impact of \$400-660 billion annually. The Boston Consulting Group (Ratajczak et al., 2023) further reports that CMOs who have adopted generative AI for content creation see measurable gains in speed-to-market and creative output yet struggle with brand consistency and quality control. This dual challenge maximizing predicted engagement while preserving brand identity represents a significant and timely problem in AI-driven marketing.
- The marketing context for this project centers on digital advertising across social media platforms such as Instagram, Twitter, Reddit, and Facebook, targeting diverse consumer audiences at multiple stages of the campaign lifecycle (launch, sustain, and post-launch phases). The target audience encompasses marketing professionals, brand managers, and content strategists at mid-to-large enterprises who manage multi-platform campaigns for well-known consumer brands. The project specifically examines three representative brand archetypes like Samsung (innovative, tech-forward, premium), Duolingo (witty, playful, gamified), and Nike (action-oriented, inspiring, bold) each with distinct voice profiles, style rules, and reference and copies captured in a structured brand reference document. The marketing outcome to improve is dual: (1) predicted CTR as a proxy for ad engagement performance, and (2) brand voice alignment score as a measure of tonal consistency. The customer journey stage most relevant to this project is the awareness-and-consideration phase, where ad copy first encounters the consumer and must simultaneously capture attention (performance) and reinforce brand identity (alignment).
- The primary research questions guiding this project are: (1) Can NLP-derived textual features of social media ad copy such as word count, question marks, exclamation points, and urgency keywords reliably predict engagement rates (CTR proxy)? (2) How effectively can sentence-level embedding models (Sentence-BERT) quantify the alignment between generated ad copy and a brand's established voice? (3) Is it feasible to build an integrated pipeline that generates multi-style and copy variants via large language models, scores each variant on both predicted performance and brand alignment, and automatically rewrites low-alignment variants to improve tonal consistency without sacrificing predicted engagement? Reviewing existing academic and industry literature is essential to situate these questions within the broader landscape of AI-driven marketing, to identify proven methodologies for CTR prediction, text style transfer, and brand personality modeling, and to discover gaps that this capstone project can address.

## 2. Organization

- The literature reviewed for this project is organized thematically into four major streams that directly map to the components of the OnBrand AdCopy pipeline. The first theme is AI-powered ad copy generation, covering how large language models and controllable text generation methods produce marketing content. The second theme is CTR prediction and advertising optimization, examining machine learning approaches to estimating click-through rates from textual and contextual features. The third theme is text style transfer and brand voice alignment, reviewing methods that adapt generated text to match a target style or tone. The fourth theme is the broader landscape of generative AI adoption in marketing, drawing on industry reports that contextualize the managerial and strategic implications of these technologies. This thematic organization reflects the integrated nature of the project, which combines generation, prediction, alignment, and optimization in a single decision-support system.
- The themes covered span several areas: advertising optimization (CTR prediction models, ad targeting), content generation (LLM-based copywriting, style transfer), sentiment analysis (SBERT-based sentiment and brand personality embedding), social media analytics (platform-level engagement patterns), and personalization (brand-specific tone adaptation). These themes collectively address the end-to-end challenge of producing ad copy that is both high-performing and brand-consistent, which is the central premise of the OnBrand AdCopy project.

## 3. Summary and Synthesis

- Several key studies inform the ad copy generation component of this project. Chai et al. (2022) propose FAST (Feedback Aware Self-Training), a method from Microsoft that improves controllability in text generation by incorporating quality feedback into the self-training loop; this is directly relevant to the auto-correction cycle in the OnBrand pipeline, where generated copy that fails brand alignment is rewritten and re-evaluated. The GCOF framework (arXiv 2402.13667, 2024) introduces a self-iterative text generation approach specifically designed for copywriting with LLMs, demonstrating that iterative refinement significantly improves both fluency and relevance of marketing text. The large-scale study (arXiv 2506.17863, 2025) examines how LLMs can be deployed for customized marketing content generation and evaluation at scale, providing evidence that LLM-generated ad copy can rival human-written alternatives when properly constrained and evaluated. For CTR prediction, Bai et al. (2025) provide a comprehensive survey of advertising CTR prediction algorithms in the Knowledge Engineering Review, cataloging approaches from logistic regression through deep learning and highlighting that gradient boosting methods remain competitive for tabular feature-based prediction. Lin et al. (2024) introduce ClickPrompt, demonstrating that CTR models can serve as strong prompt generators for adapting language models to CTR prediction finding bridges the generation and prediction components of our pipeline. Ebadi Jokandan et al. (2022) address targeted advertising in social media using hybrid convolutional learning methods. For style transfer and brand alignment, the survey (arXiv 2407.16737, 2024) provides a thorough overview of text style transfer methods, including their ethical implications. La Quatra et al. (2024) propose a self-supervised approach using Cycle-Consistent Adversarial Networks for text style transfer. Hassan et al. (2023) develop a model for brand personality word embedding, offering a quantitative framework for representing and comparing brand voice attributes in vector space. Reisenbichler et al. (2023) investigate applying LLMs to sponsored search advertising, demonstrating the practical potential of LLM-generated ad content in real marketing settings.
- Comparing these sources reveals important methodological diversity. For CTR prediction, Bai et al. (2025) survey approaches ranging from traditional feature engineering with gradient boosting to deep CTR models (DeepFM, DIN), while Maulina and Hendrawan (2026) specifically compare Gradient Boosting, XGBoost, CNN, and SVR for social media promotion analysis, finding that ensemble methods generally outperform deep models on structured feature data finding consistent with our choice of XGBoost for the performance scoring module. The Chinese LLM study (Electronic Commerce Research, 2025) and Xiao et al. (2025) both evaluate LLM-generated advertising content against human benchmarks but differ in domain and evaluation

methodology; both conclude that LLM output can match or approach human quality but requires domain-specific fine-tuning or constraint. For text style transfer, the arXiv survey (2024) and La Quatra et al. (2024) represent complementary perspectives: the former provides breadth across methods (VAE-based, GAN-based, prompting-based), while the latter offers depth on CycleGAN-based approaches. Industry reports (McKinsey, 2023; BCG, 2023; State of AI in Marketing, 2026) consistently emphasize the transformative potential of generative AI in marketing while highlighting challenges around brand consistency and quality control. Leonid (2025) demonstrates SBERT's effectiveness for sentiment-based market analysis, supporting our choice of Sentence-BERT as the embedding backbone for brand alignment scoring.

- Synthesizing the literature collectively, several key findings emerge. First, LLM-based ad copy generation has reached a maturity level where generated text can be competitive with human-written copy, but uncontrolled generation often produces generic or off-brand output (GCOF, 2024; arXiv 2506.17863, 2025). Second, CTR prediction from textual features is well-established, but most models treat text as a fixed input rather than a variable to be optimized the ClickPrompt framework (Lin et al., 2024) is a notable exception. Third, text style transfer techniques exist for adapting tone and formality, but they have rarely been applied specifically to brand voice alignment in advertising contexts. Fourth, brand personality and voice can be represented in embedding space (Hassan et al., 2023), but no existing system integrates this representation into a closed-loop pipeline that simultaneously generates, scores, and corrects ad copy. The literature thus reveals a fragmented landscape where generation, prediction, alignment, and correction are studied in isolation, creating a clear opportunity for an integrated approach.

#### **4. Marketing Relevance and Gap**

- Current AI approaches in digital marketing have made significant advances in isolated components: LLMs can generate fluent and creative ad copy (GCOF, 2024; Reisenbichler et al., 2023); gradient boosting and deep learning models can predict CTR with reasonable accuracy (Bai et al., 2025; Maulina & Hendrawan, 2026); and embedding models like SBERT can capture semantic similarity for brand personality assessment (Hassan et al., 2023; Leonid, 2025). However, a critical gap remains: no existing system integrates these capabilities into a unified pipeline that jointly optimizes both predicted performance and brand voice alignment. The standard industry practice is to treat performance optimization and brand consistency as separate, often conflicting, processes. Furthermore, the autocorrection of high-potential but off-brand copy is entirely unexplored in the academic literature; existing style transfer work focuses on generic tone dimensions rather than brand-specific voice profiles.
- The OnBrand AdCopy project directly responds to this gap by proposing an integrated decision-support platform that unifies generation, prediction, alignment, and correction in a single pipeline. Specifically, the project introduces three novel contributions: (1) a composite scoring framework that combines XGBoost-based CTR prediction with Sentence-BERT-based brand alignment into a single, user-adjustable metric; (2) an automatic correction loop where high-CTR but low-brand-alignment variants are rewritten by the LLM to match the brand's reference texts and then re-scored, reporting any performance trade-off; and (3) a structured brand reference architecture (demonstrated with Samsung, Duolingo, and Nike) that encodes tone category, voice summary, style rules, and reference ad copies in a machine-readable format for systematic alignment scoring. This integration transforms brand voice compliance from a subjective, manual process into a measurable, automated system.

#### **5. Conclusion**

- The literature review reveals that the three core capabilities required by the OnBrand AdCopy project LLM-based ad copy generation, ML-based CTR prediction, and embedding-based brand voice alignment are each individually well-supported by existing research. LLM copywriting has demonstrated human-competitive quality when properly constrained (GCOF, 2024; arXiv 2506.17863, 2025). Gradient boosting models remain state-of-the-art for CTR prediction on structured feature data (Bai et al., 2025; Maulina & Hendrawan, 2026).

Sentence-level embeddings effectively capture stylistic and semantic similarity relevant to brand personality (Hassan et al., 2023; Leonid, 2025). Industry reports confirm strong demand for AI-powered content creation with brand safety guardrails (McKinsey, 2023; BCG, 2023; State of AI in Marketing, 2026). The key takeaway is that all necessary building blocks exist, but they have not been integrated into a cohesive system.

- The proposed OnBrand AdCopy project can contribute to marketing practice by providing a decision-support tool that reduces the time and cost of ad copy creation while improving both predicted engagement and brand consistency. For marketing teams, this means faster time-to-market campaigns with higher-quality creative output. For brand managers, it provides an auditable, quantitative measure of brand voice alignment replacing subjective editorial review with a reproducible score. The composite scoring framework enables explicit trade-off analysis between performance and brand fit, allowing marketers to make informed decisions based on their strategic priorities. If implemented successfully, this system could measurably improve click-through rates while maintaining brand equity.

## 6. Proper Citations

- [1] Chai, J., Pryzant, R., Dong, V. Y., Golobokov, K., Zhu, C., & Liu, Y. (2022). FAST: Improving Controllability for Text Generation with Feedback Aware Self-Training. arXiv:2210.03167. [2] GCOF: Self-iterative Text Generation for Copywriting Using Large Language Model (2024). arXiv:2402.13667. [3] A Survey of Text Style Transfer: Applications and Ethical Implications (2024). arXiv:2407.16737. [4] LLMs for Customized Marketing Content Generation and Evaluation at Scale (2025). arXiv:2506.17863. [5] Jiang, R., Long, R., Gu, C., & Yan, M. (2025). VisuCraft: Enhancing Large Vision-Language Models for Complex Visual-Guided Creative Content Generation. arXiv:2508.02890. [6] Lin, J., Chen, B. et al. (2024). ClickPrompt: CTR Models are Strong Prompt Generators for Adapting Language Models to CTR Prediction. ACM (DOI: 10.1145/3589334.3645396). [7] La Quatra, M., Gallipoli, G., & Cagliero, L. (2024). Self-supervised Text Style Transfer Using Cycle-Consistent Adversarial Networks. ACM (DOI: 10.1145/3678179). [8] Ebadi Jokandan, S. M., Bayat, P., & Foumani, M. F. (2022). Targeted Advertising in Social Media Platforms Using Hybrid Convolutional Learning Method. J. of Electrical and Computer Engineering. [9] Reisenbichler, M., Reutterer, T., & Schweidel, D. A. (2023). Applying Large Language Models to Sponsored Search Advertising. MSI Working Paper No. 23-136. [10] Maulina, P. & Hendrawan, A. (2026). Analysis of the Best Social Media Platforms for Promotion Using Machine Learning and RFE Feature Selection. JAIC, 10(1), 513-521. [11] Leonid, T. (2025). Research and Analysis of Market Trends with Sentimental Analysis Using SBERT. ICETET-SIP 2025. [12] Bai, J. et al. (2025). A Comprehensive Survey on Advertising Click-Through Rate Prediction Algorithm. The Knowledge Engineering Review, 40, e3, 1-38. [13] Ratajczak, D. et al. (2023). How CMOs Are Succeeding with Generative AI. Boston Consulting Group. [14] Hassan, T. et al. (2023). A Model for Brand Personality Word Embedding. Journal of Travel Research, 63(1), 44-63. [15] Chinese LLMs for Advertising Slogans (2025). Electronic Commerce Research (DOI: 10.1007/s10660-025-09990-2). [16] Xiao, L., Li, X., & Ge, J. (2025). An Exploratory Research on Using Generative AI to Generate Print Ads. Customer Needs and Solutions, 12:1 (DOI: 10.1007/s40547-024-00152-3). [17] McKinsey Global Institute (2023). The Economic Potential of Generative AI: The Next Productivity Frontier. [18] The State of AI in Marketing 2026. Southeast Asia Research Report, June 2026.

# Preparing Your Data Research

## 1. Introduction

- The primary marketing objective of the OnBrand AdCopy project is to improve the efficiency and effectiveness of digital advertising copy creation by predicting engagement performance (click-through rate) and assessing brand voice alignment before campaign launch. To investigate this objective, three types of data are needed: (1) social media engagement data with textual content and measurable performance metrics (impressions, likes, shares, comments) to train the CTR prediction model; (2) structured brand reference data encoding the voice, tone, and style rules of target brands for the alignment scoring module; and (3) NLP-derived features extracted from the text content to serve as predictive inputs. The combination of these data sources enables the system to learn the relationship between textual characteristics and engagement outcomes while simultaneously establishing brand-specific benchmarks for voice consistency.
- The selected dataset the Social Media Engagement Dataset from Kaggle provides the necessary combination of textual ad content and quantitative engagement metrics to answer the project's core research questions. The dataset includes text\_content (the raw post text), along with engagement metrics (likes\_count, shares\_count, comments\_count, impressions) that enable construction of a CTR proxy metric. It also includes contextual variables such as platform, sentiment\_score, emotion\_type, and brand\_name that allow for brand-specific and platform-specific analysis. For the brand alignment component, a curated brand reference JSON file provides reference ad copies, tone categories, and style rules for three representative brands (Samsung, Duolingo, Nike), enabling Sentence-BERT-based cosine similarity scoring.

## 2. Data Source and Scope

- The primary dataset is the "Social Media Engagement Dataset" sourced from Kaggle, a publicly available third-party dataset accessed via direct CSV download. The dataset contains 12,000 records across 28 columns, covering the period of 2024. Each record represents a social media post with the following key variables: post\_id (unique identifier), timestamp, day\_of\_week, platform (Instagram, Twitter, Reddit, Facebook), user\_id, location, language (10 languages), text\_content, hashtags, mentions, keywords, topic\_category, sentiment\_score (-1 to 1), sentiment\_label, emotion\_type, toxicity\_score (0 to 1), likes\_count (0-5,000), shares\_count (0-2,000), comments\_count (0-1,000), impressions (130-99,997), engagement\_rate, brand\_name, product\_name, campaign\_name, campaign\_phase, user\_past\_sentiment\_avg, user\_engagement\_growth, and buzz\_change\_rate. The granularity is at the individual post level. After filtering for English-language posts (language == "en") and removing records with zero impressions, the working dataset contains approximately 1,197 English-language records. A supplementary brand reference JSON file provides structured voice profiles for Samsung, Duolingo, and Nike.
- The Social Media Engagement Dataset is a third-party, publicly available dataset hosted on Kaggle. It is synthetic/simulated data designed to mimic real social media engagement patterns, which makes it appropriate for developing and demonstrating the analytical pipeline without privacy or consent concerns associated with real user data. The brand reference JSON is first-party data specifically curated for this project, containing real brand ad copies and voice descriptions sourced from public brand communications (websites, social media campaigns). This combination of synthetic engagement data and real brand reference material provides a representative testbed for the OnBrand AdCopy system.
- This dataset was selected because it uniquely combines textual content with quantitative engagement metrics across multiple platforms and brands the exact combination needed to train a text-to-CTR prediction model. The text\_content field provides the raw material for NLP feature engineering (word count, character count, punctuation patterns, urgency keywords), while the engagement variables (likes, shares, comments, impressions) enable construction of a CTR proxy metric. The inclusion of brand\_name allows brand-specific analysis, and the multi-platform coverage (Instagram, Twitter, Reddit, Facebook) ensures findings generalize

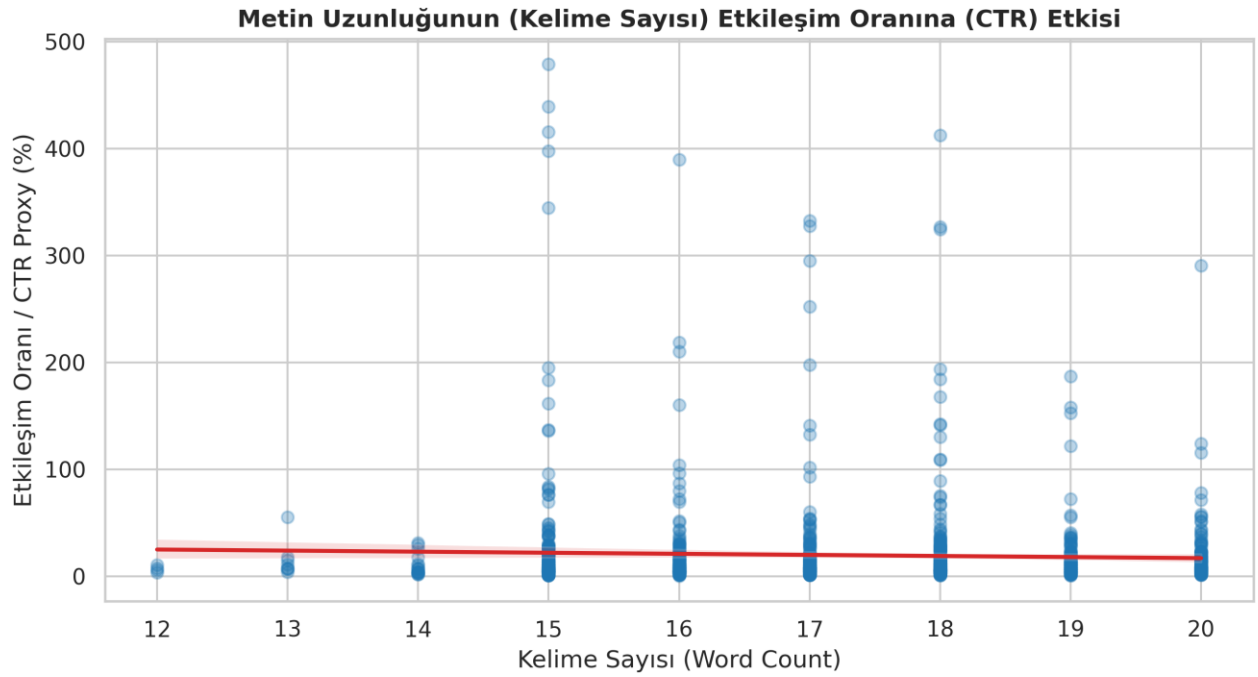
across channels. The brand reference JSON connects this engagement data to the alignment scoring module by providing the target brand voices against which generated copy will be evaluated.

### 3. Data Quality, Privacy, and Limitations

- Several data quality considerations were identified and addressed during preprocessing. First, the dataset contains posts in 10 languages; to ensure linguistic consistency for NLP feature extraction, only English-language posts were retained ( $n = 1,197$  out of 12,000), reducing the sample size but improving feature reliability. Second, records with zero impressions were excluded to avoid division-by-zero errors in CTR proxy calculation. Third, the derived CTR proxy metric exhibited extreme outliers (values exceeding 400%), likely attributable to posts with very low impressions and disproportionately high engagement; these were clipped at the 99th percentile to prevent skewing of regression analysis and visualization. Fourth, the dataset appears to be synthetically generated, as evidenced by suspicious uniform distributions (e.g., `sentiment_score` is nearly uniformly distributed between -1 and 1). While the synthetic nature limits generalizability to real campaign data, it is adequate for demonstrating the analytical pipeline and feature engineering approach.
- The Social Media Engagement Dataset, being a synthetic/simulated dataset from Kaggle, does not contain real personal identifiable information (PII) or actual user data. The `user_id` field is anonymized by design (e.g., "user\_52nwb0a6"), and location data represents city-level geography rather than precise coordinates. No consent, de-identification, or IRB approval is required for this dataset. The brand reference JSON contains only publicly available marketing copy from brand websites and social media information that is in the public domain. In the production deployment of the OnBrand AdCopy system, however, responsible use policies would need to address data governance for any real customer engagement data, ensure generated ad copy complies with advertising regulations, and implement human oversight for auto-corrected content.
- Several limitations may affect the project's scope and generalizability. First, the English-language subset (approximately 1,197 records) is relatively small for training a robust CTR prediction model; in practice, production systems would require orders of magnitude more data. Second, the synthetic nature of the data means that the learned feature-CTR relationships may not perfectly reflect real-world advertising dynamics. Third, the brand reference JSON currently covers only three brands (Samsung, Duolingo, Nike). Fourth, the dataset's engagement metrics are not true click-through rates but rather a proxy calculated from likes, shares, and comments divided by impressions; this proxy captures social engagement rather than actual ad click behavior. Fifth, the dataset lacks A/B test structure or causal design, so relationships between text features and engagement are correlational rather than causal.

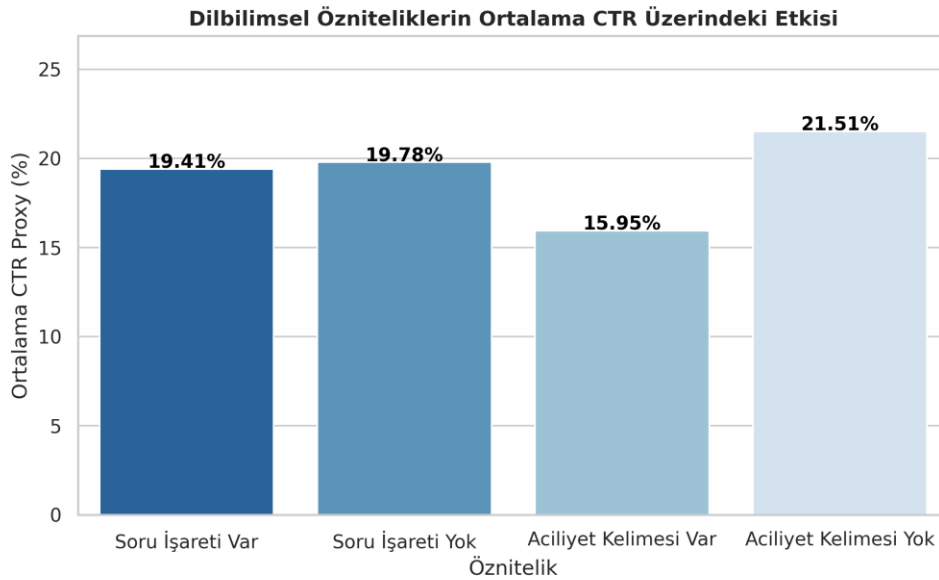
### 4. Exploratory Analysis and Marketing Insights

- Exploratory data analysis of the English-language subset ( $n = 1,197$ ) revealed several noteworthy patterns. The engineered CTR proxy metric, defined as  $((\text{likes\_count} + \text{shares\_count} + \text{comments\_count}) / \text{impressions}) \times 100$ , had a mean of approximately 19.7% with high variance and a right-skewed distribution. Text length analysis showed a mean word count of approximately 16.4 words per post. A scatter plot with regression line (Figure 1) examining the relationship between word count and CTR proxy revealed a slightly negative trend, suggesting that longer ad texts do not necessarily produce higher engagement and may in fact slightly reduce it. This finding is consistent with marketing best practices that favor concise, punchy and copy for social media platforms.



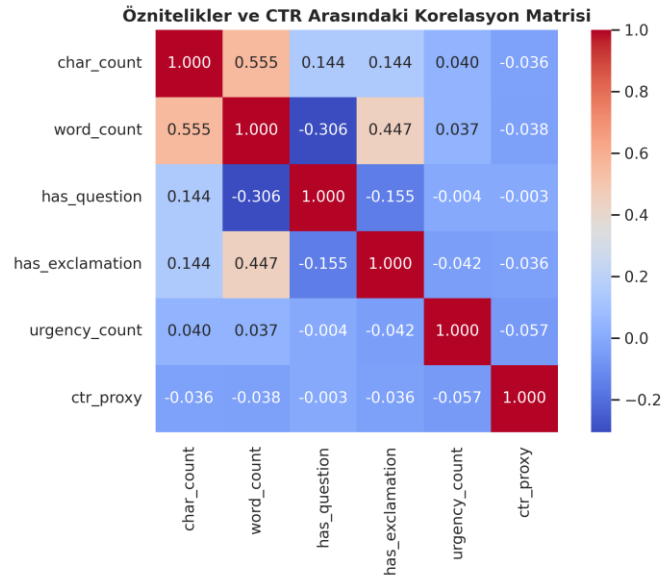
**Figure 1:** Effect of Text Length (Word Count) on Engagement Rate (CTR Proxy)

- Descriptive statistics and three key visualizations were produced to characterize the dataset. Figure 1 shows a scatter plot with OLS regression line for word count versus CTR proxy, confirming the weak negative association ( $r = -0.038$ ). Figure 2 presents a bar chart comparing mean CTR proxy across linguistic feature groups: posts containing question marks showed 19.41% average CTR versus 19.78% for those without (a negligible difference of 0.37 percentage points), while posts containing urgency keywords ("now", "today", "limited", "hurry", "exclusive", etc.) showed a markedly lower average CTR of 15.95% compared to 21.51% for posts without urgency keywords a 5.56 percentage point gap. This counter-intuitive finding suggests that explicit urgency language may reduce engagement in social media contexts, possibly due to consumer skepticism toward high-pressure tactics.



**Figure 2:** Impact of Linguistic Features on Average CTR

- Figure 3 displays a correlation heatmap of all engineered features against the CTR proxy, showing that all features have weak negative correlations with CTR (ranging from -0.003 for has\_question to -0.057 for urgency\_count), and confirming expected positive correlations between char\_count and word\_count (0.555) and between word\_count and has\_exclamation (0.447). These exploratory findings have direct implications for the marketing metrics and KPIs is central to this project. The negative association between urgency keywords and CTR proxy is particularly relevant to the OnBrand AdCopy pipeline: the CTR prediction model should assign lower predicted scores to variants that rely heavily on urgency language. The weak overall correlations between individual text features and CTR suggest that more complex, non-linear models (such as gradient boosting) are needed to capture engagement patterns that simple linear features miss justifying the choice of XGBoost over linear regression in the pipeline.



**Figure 3:** Correlation Matrix Between Features and CTR Proxy

## 5. Conclusion

- The data research confirms that the Social Media Engagement Dataset, combined with the brand reference JSON, provides a suitable foundation for the next stage of the OnBrand AdCopy capstone project. Key findings include: (1) text length has a negligible-to-slightly-negative effect on engagement; (2) urgency keywords are associated with lower CTR, a counter-intuitive result with practical implications for ad generation strategy; (3) individual NLP features show weak linear correlations with engagement, confirming the need for non-linear ML models; and (4) the data is clean enough for pipeline development after appropriate filtering and outlier treatment.
- The available data directly supports the overall marketing objective and planned AI approach of the OnBrand AdCopy project. The engagement metrics enable training of the XGBoost-based CTR prediction module, while the text\_content field provides the raw input for NLP feature engineering. The brand reference JSON supplies the target embeddings for Sentence-BERT-based alignment scoring. The EDA findings inform feature selection and model design: the weak linear correlations validate the choice of ensemble methods over linear models, and the counter-intuitive urgency keyword finding provides a concrete test case for the auto-correction loop.

## 6. Proper Citations

- [1] Social Media Engagement Dataset. Kaggle (<https://www.kaggle.com/>). Third-party, publicly available dataset. [2] Brand Reference JSON (brand\_reference.json). First-party curated dataset containing brand voice profiles for Samsung, Duolingo, and Nike, compiled from public brand communications.



# Preparing Your Technology Review

## 1. Introduction

- The OnBrand AdCopy project integrates multiple AI technologies across its pipeline: (1) Large Language Models (LLMs) via generative API services for multi-style ad copy generation and auto-correction; (2) Gradient Boosting (specifically XGBoost) for CTR prediction from engineered text features; (3) Sentence-BERT (SBERT) for semantic embedding and brand voice alignment scoring via cosine similarity; and (4) NLP feature engineering using Python libraries (regex, pandas) for extracting textual characteristics. This technology review evaluates each component and the rationale for its selection within the integrated pipeline.
- A systematic technology review is essential for this project because the system's value proposition simultaneously optimizing predicted performance and brand alignment depends on choosing the right tool for each component. An LLM that generates fluent but uncontrollable text would undermine the style-variant approach; a CTR model that requires deep learning infrastructure would exceed the project's scope; a brand alignment method that ignores semantic meaning would miss subtle tonal differences. The technology review ensures that each chosen approach is technically suitable for its role, compatible with the other pipeline components, and practically deployable within a marketing workflow where non-technical stakeholders (brand managers, copywriters) need interpretable outputs and adjustable parameters.

## 2. Technology Overview

- The three core technologies serve distinct but complementary roles. XGBoost is an ensemble machine learning algorithm based on gradient-boosted decision trees, known for high accuracy on tabular and structured data, built-in regularization, feature importance ranking, and fast training and inference times. In this project, XGBoost takes NLP-derived features (word count, sentiment score, urgency keyword presence, punctuation patterns) and predicts a CTR proxy score for each ad copy variant. Sentence-BERT (SBERT) is a modification of the BERT architecture that produces fixed-size sentence embeddings optimized for semantic similarity tasks. Using SBERT, the project encodes both the generated ad copy and the brand's reference texts into 384- or 768-dimensional vectors, then computes cosine similarity to yield a brand alignment score. Generative LLMs (accessed via API) provide the text generation capability given structured prompts specifying product, audience, and tone, the LLM generates multiple stylistic variants and, in the auto-correction loop, rewrites off-brand variants to better match the brand's voice.
- Each technology has established use cases in marketing and business applications. XGBoost and gradient boosting methods are widely used in digital marketing for CTR prediction (as surveyed by Bai et al., 2025), conversion rate optimization, customer churn prediction, and lead scoring; their strength lies in handling mixed feature types and providing interpretable feature importances that marketers can act upon. Sentence-BERT is used in marketing for sentiment analysis, content recommendation, customer feedback clustering, and semantic search; in brand management, it enables quantitative comparison of messaging across campaigns, channels, and time periods. LLMs are increasingly deployed for ad copy generation, email personalization, product description writing, chatbot content, and social media content creation; the BCG report (Ratajczak et al., 2023) found that 70% of CMOs are either piloting or scaling generative AI for content creation.

## 3. Relevance to Your Marketing Project

- Each technology directly addresses a specific component of the OnBrand AdCopy pipeline. XGBoost predicts CTR proxy scores from engineered text features, enabling the system to rank and copy variants by expected engagement performance before launching this supports the KPI of click-through rate improvement. SBERT computes brand alignment scores by measuring the semantic similarity between generated copy and the brand's approved reference texts, directly supporting the KPI of brand voice consistency. The generative LLM produces the ad copy variants themselves and performs the auto-correction rewriting, enabling rapid creative

output at scale. Together, these technologies automate the two decisions that currently require significant human effort: "Which copy will perform best?" (CTR prediction) and "Does this copy sound like our brand?" (alignment scoring).

- In the marketing workflow, the OnBrand AdCopy system fits primarily in the campaign planning and content creation stages. The typical workflow is: (1) a marketing manager specifies the product, target audience, and desired tone; (2) the LLM generates 4-5 stylistic variants; (3) XGBoost scores each variant for predicted performance; (4) SBERT scores each variant for brand alignment; (5) the composite score ranks the variants; (6) off-brand variants enter the auto-correction loop for rewriting and re-scoring. This fits between the strategy/briefing stage (upstream) and the media buying/A/B testing stage (downstream), reducing the number of variants that need to be manually reviewed and tested.

#### 4. Comparison and Evaluation

- For CTR prediction, three candidate approaches were evaluated: (1) Logistic Regression / Linear Models simple and interpretable but limited in capturing non-linear feature interactions, which the EDA showed are important (weak linear correlations but potentially stronger non-linear patterns); (2) XGBoost / Gradient Boosting handles mixed feature types, captures non-linear interactions, provides feature importance, and trains quickly on moderate-sized datasets; (3) Deep CTR Models (DeepFM, DIN) powerful but require large datasets, GPU infrastructure, and offer less interpretability. XGBoost was selected as the best fit for this project: it balances accuracy with interpretability, runs efficiently on the available dataset size (~1,197 English records), and its feature importance output directly informs the generation module about which text characteristics drive engagement.
- For brand alignment scoring, two primary approaches were considered: (1) Sentence-BERT with cosine similarity produces dense semantic embeddings, supports fast batch computation, captures meaning beyond lexical overlap, and is available via the sentence-transformers Python library with no API cost; (2) LLM-based scoring via prompting potentially more nuanced but slower, more expensive (API calls per variant), and less reproducible due to prompt sensitivity. SBERT was selected for its speed, cost-effectiveness, reproducibility, and proven performance in semantic similarity tasks (Leonid, 2025). The SBERT alignment score is supplemented by rule-based heuristics (formality level, sentence length distribution, vocabulary overlap with brand style rules) to capture structural brand voice dimensions that pure semantic similarity may miss.
- The evaluation criteria used to compare technologies are tailored to the marketing task. For CTR prediction, the relevant metrics are prediction accuracy (RMSE, MAE on held-out data), feature interpretability (can marketers understand why a variant score high/low?), and inference latency (can the model score 5 variants in under a second?). For brand alignment, the criteria are semantic fidelity (does the score reflect human judgment of brand consistency?), discrimination ability (can it distinguish on-brand from off-brand copy?), and scalability (can it process batch evaluations efficiently?). For text generation, the criteria are output diversity (does it produce meaningfully different stylistic variants?), controllability (does it follow the tone instructions?), and correction quality (does rewriting improve alignment without destroying the original message?). These marketing-specific criteria ensure that the chosen technologies serve the business objective, not just technical benchmarks.

#### 5. Use Cases and Examples

- Several real-world marketing applications demonstrate the viability of the technologies used in this project. Reisenbichler et al. (2023) report that applying LLMs to sponsored search advertising at a major European retailer improved ad relevance scores and reduced manual copywriting time by over 50%. The ClickPrompt system (Lin et al., 2024) demonstrated at Huawei that CTR prediction models can be used to guide LLM prompting for ad content, achieving improvements in both prediction accuracy and content relevance. The BCG report (2023) cites case studies from consumer goods companies where generative AI reduced campaign

content creation time from weeks to days while maintaining brand guidelines through structured prompt engineering. The State of AI in Marketing 2026 report documents that leading Southeast Asian e-commerce platforms now use SBERT-based content matching systems to ensure auto-generated product descriptions align with seller brand profiles.

- The most instructive example for the OnBrand AdCopy project is the ClickPrompt system (Lin et al., 2024), which demonstrates that CTR prediction models and language models can be effectively coupled: the CTR model provides signal about what content features drive clicks, and this signal guides the LLM's generation process. This example validates the core architecture of our pipeline. Similarly, Hassan et al. (2023) show that brand personality can be meaningfully captured in embedding space and used for quantitative comparison supporting our use of SBERT for brand alignment. The key lesson from these examples is that integration of prediction and generation yields better outcomes than using either in isolation.

## 6. Limitations, Risks, and Opportunities

- Several limitations and risks are relevant to the chosen technology stack. For LLM-based generation, key risks include hallucination (generating factually incorrect product claims), brand safety (producing content that could offend or misrepresent the brand), and prompt injection (adversarial inputs causing unintended outputs). For XGBoost-based CTR prediction, the primary limitation is that the model learns correlational patterns from historical data and may not generalize to novel campaign contexts or emerging platforms. For SBERT-based alignment scoring, the risk is that cosine similarity captures semantic meaning but may miss pragmatic dimensions of brand voice (humor, irony, cultural references) that are context-dependent. Additionally, the auto-correction loop introduces a dependency: if the LLM's rewrite significantly alters the message meaning, the CTR prediction may no longer be valid for the rewritten text, requiring re-scoring.
- Several opportunities exist for improving and extending the system. First, the SBERT alignment score could be augmented with few-shot LLM-based evaluation as a secondary quality check for edge cases where semantic similarity alone is insufficient. Second, the brand reference architecture could be expanded with automated brand voice profiling, where the system analyzes a brand's existing content corpus to automatically extract style rules, vocabulary preferences, and tone dimensions. Third, human-in-the-loop feedback could be integrated, where marketing managers rate the quality of generated and corrected variants, creating a reinforcement signal for fine-tuning the composite scoring weights. Fourth, the system could be extended to multilingual markets by leveraging SBERT's multilingual variants and cross-lingual generation capabilities.

## 7. Conclusion

- The technology review confirms that the selected technology stacks XGBoost for CTR prediction, Sentence-BERT for brand alignment scoring, and API-based LLMs for generation and autocorrection is well-suited for the OnBrand AdCopy project. Each technology is mature, well-documented, and has demonstrated effectiveness in marketing or closely related applications. XGBoost provides the optimal balance of accuracy and interpretability for the available dataset; SBERT offers fast, cost-effective semantic similarity computation; and LLM APIs provide state-of-the-art generation capabilities without the overhead of model training. The integration of these three components into a single pipeline represents a novel contribution, as no existing tool or platform combines CTR prediction, brand alignment scoring, and autocorrection in a unified decision-support system.
- If implemented successfully, the OnBrand AdCopy system can contribute to marketing practice in several measurable ways: (1) reducing ad copy creation time by automating variant generation and initial quality screening; (2) improving predicted click-through rates by surfacing variants with optimal feature profiles; (3) ensuring brand consistency through quantitative alignment scoring, replacing subjective editorial review; and (4) providing marketing managers with explicit trade-off visibility between performance and brand fit through the adjustable composite scoring framework. The practical value extends beyond efficiency gains: by making

brand voice compliance measurable and automated, the system addresses a long-standing gap in marketing technology that currently relies on human judgment alone.

## 8. Proper Citations

- [1] XGBoost: Chen, T. & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. KDD 2016. [2] Sentence-BERT: Reimers, N. & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. EMNLP 2019. [3] sentence-transformers Python library: <https://www.sbert.net/>. [4] scikit-learn: Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, 2825-2830. [5] Bai, J. et al. (2025). A Comprehensive Survey on Advertising CTR Prediction Algorithm. KER, 40, e3. [6] Lin, J. et al. (2024). ClickPrompt: CTR Models are Strong Prompt Generators. ACM. [7] Reisenbichler, M. et al. (2023). Applying LLMs to Sponsored Search Advertising. MSI Working Paper 23-136. [8] Ratajczak, D. et al. (2023). How CMOs Are Succeeding with Generative AI. BCG. [9] McKinsey Global Institute (2023). The Economic Potential of Generative AI. [10] Leonid, T. (2025). Market Trends with Sentimental Analysis Using SBERT. ICETET-SIP. [11] Hassan, T. et al. (2023). Brand Personality Word Embedding. JTR, 63(1), 44-63. [12] The State of AI in Marketing 2026. Southeast Asia Research Report.